Nest, Run, Fold: The Form of the BBP Digit Machine

Compilation first, two depths, a crest, a seam, a Mold, and why zero needs an inversion

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Abstract

We read the Bailey–Borwein–Plouffe formula as a machine rather than a recipe. Every term is a phase $\{16^{n-k}/(8k+j)\}$ on one circle, present from step 0 and moved by the single map $x \mapsto \{16x\}$; each hexadecimal digit of π is a difference that this motion emits. Organizing the machine by one Form (compile first, nest, run, encapsulate, invert, fold, retain, read) yields exact statements at every scope. A lane runs from step $k + \lceil \log_{16}(8k+j) \rceil - 1$, but its layer's value arrives only near $\log_{16}(64k^2/15)$; in between, the lanes move value-free carries. BBP is the only zero-sum member of the Adamchik–Wagon line of formulas, and that is exactly what makes π the maximum of its own partial integral. The whole line is the shadow of $\log(1+i) = \Sigma\, ((1+i)/2)^n/n$: π is its inversion-odd part, log 2 its fold-even part, base 16 is the base −4 formula read across two seams, and π and log 2 share one integrand kernel that changes sign at the wheel's first step. Every register obeys new = 16·old − emitted, so retaining the emitted integer reverses every step; the lane field is reversible by itself, and the hex digits of π are exactly what the projection to π erases. A lane's loop balances only if ⟨16⟩ contains an involution: the full inversion forces balance (Midy), a partial one only permits it, and none forbids it. In the wheel and in the loops alike, the inversion is not an added rule but the half-period of an even closure. The loop means alone predict the drift of the machine's carry to within 2%. Finally, the Bailey–Crandall one-point recurrence misreads digit n exactly when $\{16^n\pi\} < T_n \approx 1/(4n^2)$; it never does for $n \le 4 \times 10^6$, and a proof that it never does would require excluding zero runs of length about $\log_{16}(4n^2)$ in π, while known bounds exclude only runs longer than about 6n.

Keywords: BBP formula; digit extraction; Midy's theorem; Dirichlet characters; class numbers; natural extension; reversible dynamics; normality of π. *MSC 2020:* 11Y60, 11A63, 11K16, 11R29.

1 Introduction: the Form

The formula of Bailey, Borwein and Plouffe [3],

$$\pi = \Sigma_{k\ge 0}\, 16^{-k} \left(4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6) \right), \tag{1}$$

made it possible to compute a hexadecimal digit of π without computing the digits before it. Its usual reading is algorithmic: split the sum at k = n, reduce 16^{n-k} modulo 8k+j, add a short tail, read off a digit.

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Here we read the same formula as a machine, a single motion acting on a field of phases, and ask what form that machine has. The question is not whether new formulas for π exist but what the existing one already does. In that sense little here is new, only new to the reader, and where a result is classical we name its source.

We organize everything by one pattern, which we call the Form. *Compile first:* everything the machine uses is present at the start, and nothing is spawned later. *Nest* until nesting can no longer hold a term, then *run*. *Encapsulate:* several motions compiled into one register, whose value arrives later than its motion. *Invert* and *fold:* a structure taken against its mirror, by subtraction or by addition; the *seam* is where a reading turns over. *Retain:* keep the prior form so that every step can be undone (the Mold). *Read:* the readout is the difference a step produces, not a separate gate applied afterwards. The claim of this paper is that each of these is a theorem about the BBP machine, and that the same Form appears at every scope: the lane, the layer, the wheel behind the family of formulas, the loops of the lanes, and π's own phase.

Table 1 separates what is proved, what is verified numerically and how far, and what remains open. Every number in the paper is produced by a script listed in the Appendix.

Table 1. Results ledger.

Result	Status	Origin
Prop. 1–2. One motion: every term is a phase moved by $x \mapsto \{16x\}$; birth is not an event; each digit is a difference of crossings.	Proved; 8,004,000 crossings and 2,000 digits checked	Elementary; the reading is ours
Prop. 3. Nest, then run: lane (k, j) first crosses at step $k + \lceil \log_{16} d \rceil - 1$.	Proved; 7,988 of 7,988 lanes	Elementary
Thm. 4. Two depths: a layer's value arrives at age $\lceil \log_{16}(1/V_k) \rceil - 1$; the four-lane motion before it is value-free carry.	Proved; identity 45,030 of 45,030; onsets 10,000 of 10,000	New
Thm. 5. Crest: BBP is the only zero-sum member of the Adamchik–Wagon line, and π is the maximum of its partial integral.	Proved	Line: [1]. Crest: new
Thm. 6, Cor. 7. Wheel: the line is the shadow of log(1+i); seam; sparsest formulas are the wheel $\mp R/4$; π and log 2 share one crest kernel.	Proved	Ingredients classical [1–3]; arrangement new
Prop. 8, Thm. 9. Mold: every register is new = 16·old – emitted; the lane field is reversible; π's digits are the erased coordinate.	Proved; field run 600 steps back exactly	Construction: Fredkin [9, 12]; natural extension [11]
Thm. 10, Cor. 11. Loop lean as a character sum; class-number term.	Proved; 488 prime lanes to 2.4×10^{-14}	Method: Girstmair [7, 8]; lanes: new
Thm. 12. Zero needs an inversion: none ⇒ never balanced; full ⇒ balanced; partial ⇒ exact lean formula, either outcome.	Proved; all 512,002 lanes with $k \le 128{,}000$ classified	New

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Result	Status	Origin
Cor. 13. Lanes 8k+5 and 8k+6 never hold the full inversion, yet 1,263 of them balance ($k \le 128{,}000$).	Proved; counted	New; corrects a natural guess
Obs. 14. Loop means predict the drift of the carry C_n.	Numerical, $n < 3{,}000$: within 0.2–1.6%	New
Obs. 15. The drift grows like about $3.7\sqrt{n}\ln n$.	Numerical, $1.6 \times 10^4 \le n \le 1.28 \times 10^5$	Open
Thm. 16. The one-point traveler misreads digit n iff $\{16^n\pi\} < T_n \approx 1/(4n^2)$.	Proved; no misread for $n \le 4 \times 10^6$	Recurrence and conjecture: [5]; criterion: new
The traveler never misreads; π is normal in base 16.	Open	Needs zero-run bounds near $2\log_{16} n$

2 Compile first: one motion on one circle

Write $L_j = \Sigma_{k\ge 0}\, 16^{-k}/(8k+j)$ for j = 1, …, 8, so that (1) reads $\pi = 4L_1 - 2L_4 - L_5 - L_6$. A *lane* is a pair (k, j) with $j \in \{1, 4, 5, 6\}$, of modulus $d = 8k+j$ and weight $c_j \in \{4, -2, -1, -1\}$. A *layer* is the four lanes with the same k; its value is

$$V_k = \Sigma_j\, c_j/(8k+j) = (120k^2 + 151k + 47) \,/\, ((8k+1)(2k+1)(8k+5)(4k+3)), \tag{2}$$

so that $\pi = \Sigma_{k\ge 0}\, 16^{-k}\, V_k$. For every lane and every step $n \ge 0$ define the phase

$$x_{k,j}(n) = \{\, 16^{n-k} \,/\, (8k+j) \,\}, \tag{3}$$

where $\{\cdot\}$ is the fractional part. For $k > n$ the phase is simply $16^{n-k}/d$, smaller than 1/d: the term is already there, nested k – n levels below its own denominator.

Proposition 1 (one motion). *For every lane and every $n \ge 0$, $x_{k,j}(n+1) = T(x_{k,j}(n))$ with $T(x) = \{16x\}$. At $n = k$ the phase equals 1/d exactly ($d \ge 2$). The weighted sum $S_n = \Sigma_{k\ge 0}\, \Sigma_j\, c_j\, x_{k,j}(n)$ satisfies $S_0 = \pi - 4$ and $S_n \equiv 16^n\pi \pmod 1$.*

Proof. $\{16\{y\}\} = \{16y\}$ for every real y, and $\{1/d\} = 1/d$. Summing $c_j\, 16^{n-k}/d$ over all lanes gives $16^n\pi$, and replacing each term by its fractional part changes the sum by an integer. At n = 0 the only term with a nonzero integer part is 4/1, from the lane d = 1. ■

The machine therefore never starts a lane. The field is compiled in full at step 0, the unborn tail included in S_0, and one law moves all of it. What the usual algorithm calls the birth of a term, the moment it enters the modular part of the computation, is only the moment its phase reaches 1/d under the same law that moved it before (the dashed diagonal in Fig. 1A).

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Proposition 2 (the read is a difference). *Let $q_{k,j}(n) = \lfloor 16x_{k,j}(n) \rfloor$ be the crossing of a lane at step n, $C_n = \Sigma\, c_j\, q_{k,j}(n)$ the weighted crossing of the field, and $I_n = \lfloor S_n \rfloor$. Then $S_{n+1} = 16S_n - C_n$, and the (n+1)-th hexadecimal digit of π after the point is*

$$\delta_{n+1} = C_n + I_{n+1} - 16\, I_n. \tag{4}$$

Proof. $16x = q + T(x)$ for every phase; multiply by c_j and sum. By Proposition 1, $S_n = I_n + \{16^n\pi\}$, and $16\{16^n\pi\} = \delta_{n+1} + \{16^{n+1}\pi\}$. ■

A crossing $q = 16x - T(x)$ is the difference between the unwrapped and the wrapped motion. Nothing inspects the phase to produce it; the step emits it. The digit of π is a difference of such differences, corrected by the change in the integer part of the lift. This is the Form's "the difference is the gate" in exact terms: what we call reading a digit is our name, after the fact, for a difference the motion has already made.

We ran the one-motion machine with exact rational phases (8,160 phases, 2,000 steps) against a standard digit-extraction implementation. All 8,004,000 crossings agree, every lane sits at exactly 1/d when n = k (8,000 of 8,000), no unborn phase ever crosses, and the 2,000 digits produced are those of π. Identity (4) holds at all 2,999 steps checked up to n = 3,000.

3 Nest until you cannot, then run

Proposition 3 (activation depth). *Let $d = 8k + j \geq 2$ and $e = \lfloor \log_{16} d \rfloor$. The lane emits no crossing at steps $n < k + e - 1$, during which its phase is exactly $16^{n-k}/d$, and it emits a positive crossing at $n = k + e - 1$.*

Proof. As long as $16^{n-k+1} < d$, the phase is $16^{n-k}/d$ and sixteen times it is below 1. At $n = k + e - 1$ we have $16^{n-k+1} = 16^e \geq d$. No modulus $8k + j$ with $j \in \{1, 4, 5, 6\}$ is a power of 16 except $d = 1$, whose phase is 0 forever. ■

A lane nests, its phase growing sixteenfold per step without crossing, until nesting can no longer hold it, and then it runs. The front is the orange staircase in Fig. 1A; it was checked for every lane that crossed within 2,000 steps (7,988 of 7,988). Two features of the Form are visible here. The run is forced, not scheduled: no rule decides when a lane starts, the size of its denominator does. And before the run there is already motion, since the phase grows; the nest is accumulation, not idle storage.

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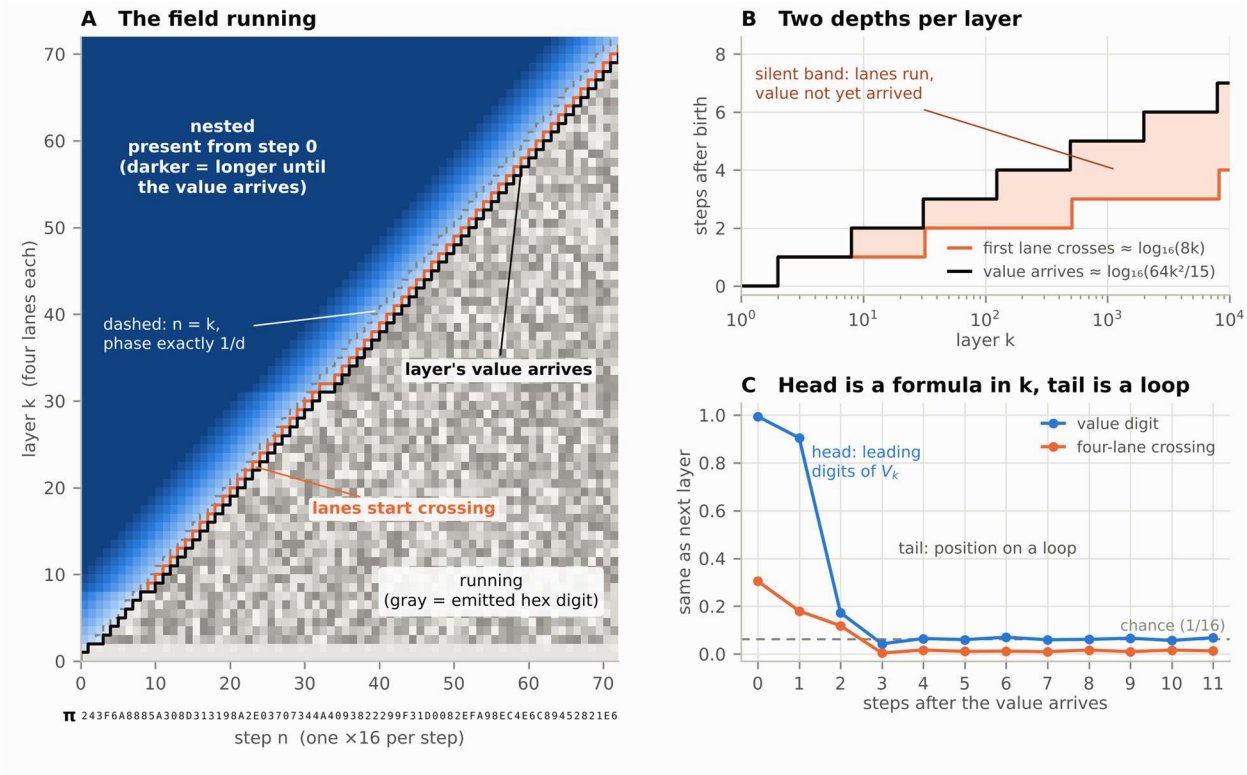


Figure 1. The machine running. (A) Layers k (rows) against steps n (columns). Blue: nested phases, present from step 0 (darker means more steps until the layer's value arrives). Gray: running phases, shaded by the hex digit they emit. Orange staircase: first lane crossing (Proposition 3). Black staircase: arrival of the layer's value (Theorem 4). The row under the panel is π, each column folded by (4). (B) The two depths for k up to 10^4. (C) Consecutive layers share the head of their value digits, a formula in k, and nothing after it, a position on a loop.

4 Encapsulate: the layer and its second depth

The four weights of a layer satisfy $\Sigma_j c_j = 0$ and $\Sigma_j j\, c_j = -15$, so

$$V_k = 15/(64k^2) + O(k^{-3}). \tag{5}$$

Each of the four phases is about 1/(8k) at birth, but their compiled value is two orders of magnitude smaller. The layer is a register into which four motions are encapsulated, and its value arrives later than its motion.

Theorem 4 (two depths and a value-free carry). *Let $k \geq 1$. For $m \geq 0$ let $w_k(m) = \lfloor 16\{16^m V_k\} \rfloor$ be the (m+1)-th hex digit of V_k, let $H_k(m) = \Sigma_j c_j q_{k,j}(k+m)$ be the layer's signed crossing at age m, and let $J_k(m) = \lfloor 16^m V_k \rfloor - \Sigma_j c_j \lfloor 16^m/(8k+j) \rfloor$.*

(a) The first nonzero value digit occurs at age $\lfloor \log_{16}(1/V_k) \rfloor - 1$, about $\log_{16}(64k^2/15) - 1$, whereas the first lane crosses at age $\lfloor \log_{16}(8k+1) \rfloor - 1$.

(b) $J_k(m) = \lfloor \Sigma_j c_j x_{k,j}(k+m) \rfloor \in \{-4, \ldots, 3\}$, $J_k(0) = 0$, and

$$H_k(m) = w_k(m) + 16\, J_k(m) - J_k(m+1). \tag{6}$$

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(c) Consequently $\Sigma_{m\ge 0} H_k(m)\, 16^{-m-1} = V_k$: the four-lane signal differs from the layer's digits by a carry flow whose total value is zero.

Proof. (a) is the position of the first nonzero hex digit of $V_k < 1$, with (5), and Proposition 3 for the lanes. (b) Write $y_j = 16^m/(8k+j)$. Then $q_{k,j}(k+m) = \lfloor 16 y_j \rfloor - 16\lfloor y_j \rfloor$ and $w_k(m) = \lfloor 16^{m+1} V_k \rfloor - 16\lfloor 16^m V_k \rfloor$, and (6) follows by substitution. Since $16^m V_k = \Sigma\, c_j\, y_j$ and $\Sigma\, c_j \lfloor y_j \rfloor$ is an integer, $J_k(m) = \lfloor \Sigma\, c_j \{y_j\} \rfloor$, which lies in $\{-4, \ldots, 3\}$ because the positive weights sum to 4 and the negative ones to −4. (c) Multiply (6) by 16^{-m-1} and sum; the carry terms telescope to $J_k(0) = 0$. ■

For $k \le 10^4$ the onset in (a) holds for every layer (10,000 of 10,000), and the band in which the lanes run while the layer is silent in value is 3 steps wide in 78% of layers and 2 steps in 18%. Identity (6) was checked at 45,030 ages ($k \le 1{,}500$, $m < 30$). The carry dominates the motion: over 40 active ages of each layer $k \le 2{,}000$ the four-lane signal H has variance 484, the value digits w only 21.4. The first nonzero H coincides with the value onset in 69.5% of layers, precedes it in 12.0% and follows it in 18.5%; the carries jitter the visible onset but cannot move value. Consecutive layers ($k = 1{,}000$ to $3{,}000$), aligned at their value onset, share their first value digit 99% of the time, their second 90%, their third 17%, and afterwards only chance, 1/16 (Fig. 1C). The head of a layer is a formula in k. Its tail is a position on a loop whose length, the order of 16 modulo the reduced denominator of V_k, runs 1, 3, 30, 105, 45, 165, 819, 45, 66 for $k = 0, \ldots, 8$ with no evident rule.

Encapsulation needs the inversion in the weights. If $\Sigma_j c_j$ were not zero, the layer value would be of order $\Sigma c/(8k)$, as large as its lanes, and the silent band would close. Section 5 shows that among the BBP-type formulas for π this inversion singles out BBP.

5 Invert: the crest

BBP is one point on a line of formulas found by Adamchik and Wagon [1]: for every real r,

$$\pi = \Sigma_{k\ge 0}\, 16^{-k} \Sigma_{j=1}^{8} c_j(r)/(8k+j), \quad c(r) = b - rR, \tag{7}$$

with b = (4, 0, 0, −2, −1, −1, 0, 0) and R = (−8, 8, 4, 8, 2, 2, −1, 0). The vector R is the zero relation $\Sigma_j R_j L_j = 0$ [2]. Since $\int_0^{1/\sqrt{2}} x^{8k+j-1}\, dx = 2^{-j/2}\, 16^{-k}/(8k+j)$,

$$\Sigma_j c_j L_j = \int_0^{1/\sqrt{2}} W_c(x)/(1 - x^8)\, dx, \quad W_c(x) = \Sigma_j c_j\, 2^{j/2}\, x^{j-1}, \quad W_c(1/\sqrt{2}) = \sqrt{2}\, \Sigma_j c_j. \tag{8}$$

For BBP the numerator is $W(x) = 4\sqrt{2} - 8x^3 - 4\sqrt{2}x^4 - 8x^5$, the integrand from which the formula was first derived [3]. It factors as

$$W(x) = -8\,(x - 1/\sqrt{2})(x^2 + 1)(x^2 + \sqrt{2}x + 1), \tag{9}$$

so its only real zero is $1/\sqrt{2}$.

Theorem 5 (crest). *On the line (7) the following are equivalent: (i) $r = 0$; (ii) $\Sigma_j c_j(r) = 0$; (iii) $a = 1/\sqrt{2}$ is a critical point of $F_r(a) = \int_0^a W_{c(r)}(x)/(1 - x^8)\, dx$; (iv) $V_k(r) = O(k^{-2})$. For $r = 0$,*

$$\pi = \max_{0\le a<1} \int_0^a W(x)/(1 - x^8)\, dx, \quad \text{attained only at } a = 1/\sqrt{2}. \tag{10}$$

In the variable $y = \sqrt{2}x$ this reads $\pi = \max_{0\le b<\sqrt{2}} \int_0^b 16(y-1)/(y^4 - 2y^3 + 4y - 4)\, dy$, attained at $b = 1$, the root of the numerator.

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Proof. $\Sigma_j c_j(r) = -15r$ gives (i)⇔(ii). $F_r'(1/\sqrt{2}) = (16/15)\sqrt{2}\,\Sigma_j c_j(r)$ gives (ii)⇔(iii). $V_k(r) = \Sigma_j c_j(r)/(8k) + O(k^{-2})$ gives (ii)⇔(iv). For r = 0, (9) shows W > 0 on [0, 1/√2) and W < 0 on (1/√2, 1), while $1 - x^8 > 0$; so F_0 rises up to 1/√2 and falls afterwards. ■

The integral that produced BBP stops exactly where its integrand changes sign. π is not only the value of that integral but the crest of its running total, and the condition that puts the endpoint on the crest is the same zero-sum inversion that makes the four lanes cancel at leading order and opens the silent band of Section 4. We have not seen this characterization stated; it is elementary, and we offer it as an instance of the Form rather than as a deep fact. Section 6 shows where the crest point comes from.

6 Fold and seam: the wheel behind the line

Let s = (1 + i)/2. One step multiplies by s: a turn of 45° and a shrink by 1/√2. Then $s^4 = -1/4$, $s^8 = 1/16$ and $s^{-1} = 1 - i = 2\bar{s}$, where $\bar{s}$ is the mirror (complex conjugate) of s.

Theorem 6 (the wheel). *(a)* $\Sigma_{n\geq 1} s^n/n = \log(1+i) = \tfrac{1}{2}\log 2 + i\pi/4$. *Grouping n = 8k + j,*

$$\pi = \Sigma_{j=1}^{8} 4\,\mathrm{Im}(s^j)\,L_j, \quad \log 2 = \Sigma_{j=1}^{8} 2\,\mathrm{Re}(s^j)\,L_j, \tag{11}$$

with π weights (2, 2, 1, 0, −½, −½, −¼, 0) = c(−¼) and log 2 weights (1, 0, −½, −½, −¼, 0, ⅛, ⅛).

(b) Fold and inversion: $\log(1+i) + \log(1-i) = \log 2$ *and* $\log(1+i) - \log(1-i) = i\pi/2$. *π is the part of the wheel that changes sign under the mirror* $s \mapsto \bar{s}$; *log 2 is what survives folding the wheel onto its mirror.*

(c) Seam: $s^{j+4} = -\tfrac{1}{4}\,s^j$, *so the π weights satisfy* $w_{j+4} = -\tfrac{1}{4}\,w_j$. *Grouping n = 4k + j instead gives the base −4 formula* $\pi = \Sigma_{k\geq 0}(-\tfrac{1}{4})^k\,(2/(4k+1) + 2/(4k+2) + 1/(4k+3))$ *[1, 2]; the base-16 formulas are this one read across two seams.*

(d) Sparsity: c(r) has four nonzero weights exactly at r = 0 (lanes 1, 4, 5, 6: BBP) and at r = −½ (lanes 2, 3, 4, 7: $2\pi = 8L_2 + 4L_3 + 4L_4 - L_7$*), six at r = −¼ (the wheel) and seven otherwise. The two sparsest formulas are the wheel* $\mp R/4$, *and BBP is the one with zero sum.*

(e) The run unwinds the nest: one machine step multiplies by $16 = (1-i)^8 = s^{-8}$, *eight inverse wheel steps.*

Proof. (a) $-\log(1-s) = \log 2 - \log(1-i)$, and $s^n = s^j\,16^{-k}$. (b) $(1+i)(1-i) = 2$ and $(1+i)/(1-i) = i$. (c) $s^4 = -\tfrac{1}{4}$. (d) The entries of c(r) are 4 + 8r, −8r, −4r, −2 − 8r, −1 − 2r, −1 − 2r, r, 0. (e) $(1-i)^2 = -2i$. ■

In the coordinates of the Form, X (the real axis) accumulates log 2, Y (the imaginary axis) accumulates π, and Z, the radial factor 1/√2 per step, is encapsulation, closing a layer every eight steps at 1/16. Their negatives are the mirror $\bar{s}$ (−Y) and the inverse step 1 − i (−Z), which is the run. The seam is where the reading turns over: the face j = 1, 2, 3 is read positively, the seam j = 4 contributes nothing to π, the face j = 5, 6, 7 is the first face turned over and shrunk by 4, and j = 8 closes the turn on the positive axis. One crossing of the seam gives the alternating base −4 reading; two crossings restore the face, and that is base 16 (Fig. 2A).

Corollary 7 (fold to unity; one kernel for both crests). $s + \bar{s} = 1$ *and s is perpendicular to* $\bar{s}$, *so* $\Sigma\, s^n/n = -\log(1-s) = -\log \bar{s}$: *the X and Y shadows read off the shrink and the turn of the mirror step. One full turn sums to* $\Sigma_{j=1}^{8} s^j = (15/16)\,i$, *purely imaginary, so the log 2 weights in (11) sum to zero while the π weights sum to 15/4. With*

$$K(x) = 2\,(1/\sqrt{2} - x)\,/\,|1 - e^{i\pi/4}\,x|^2,$$

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$$\log 2 = \int_0^{1/\sqrt{2}} K(x)\,dx, \qquad \pi = \int_0^{1/\sqrt{2}} 4K(x)/(1-x^2)\,dx, \tag{12}$$

where the second integral is BBP's. K changes sign exactly at $x = 1/\sqrt{2}$, the point where the ray toward $e^{i\pi/4}$ passes closest to 1, which is the wheel's first step s. Hence $\log 2 = \max_{a\geq 0}(-2\ln|1 - e^{i\pi/4}a|)$, and both constants are crests at the same point. The wheel's own π formula has weight sum 15/4 and is not on a crest; adding $-R/4$ puts it there, and the result is BBP.

Proof. $s/\bar{s} = i$ gives the turn sum $s(1-s^8)/(1-s) = (15/16)\cdot s/\bar{s}$. The X numerator is $W_X(x) = 2(x^4-1)(x-1/\sqrt{2})(x^2+\sqrt{2}x+1)$; with $1-x^8 = (1-x^4)(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)$ and $x^2-\sqrt{2}x+1 = |1-e^{i\pi/4}x|^2$ this gives $W_X/(1-x^8) = K$, an exact derivative of $-\ln|1-e^{i\pi/4}x|^2$. Dividing (9) by $1-x^8$ in the same way gives $W/(1-x^8) = 4K/(1-x^2)$. The foot of the perpendicular from 1 to the line through $e^{i\pi/4}$ is $(1/\sqrt{2})e^{i\pi/4} = s$. ■

This is the geometric origin of the crest in Theorem 5: π and log 2 are integrals of the same kernel along the ray from 0 to s, weighted by $4/(1-x^2)$ and by 1, and the kernel vanishes at s because s and $1-s = \bar{s}$ are perpendicular. The fold lands on its crest by itself (X cancels over a full turn); the inversion needs the zero relation to get there.

Written in components, $s = ½ + ½i$, and the fold of the step onto its mirror is $s + \bar{s} = (½ + ½) + (½ - ½)i = 1$. On X the two halves reinforce into unity; on Y they cancel to zero. π lives on the axis where the fold reads zero, which is why it is the inversion-odd part of the wheel, and log 2 on the axis where the fold reads unity. The two readouts the Form distinguishes, unity from a fold and zero from an inversion, are the two components of one step taken against its mirror.

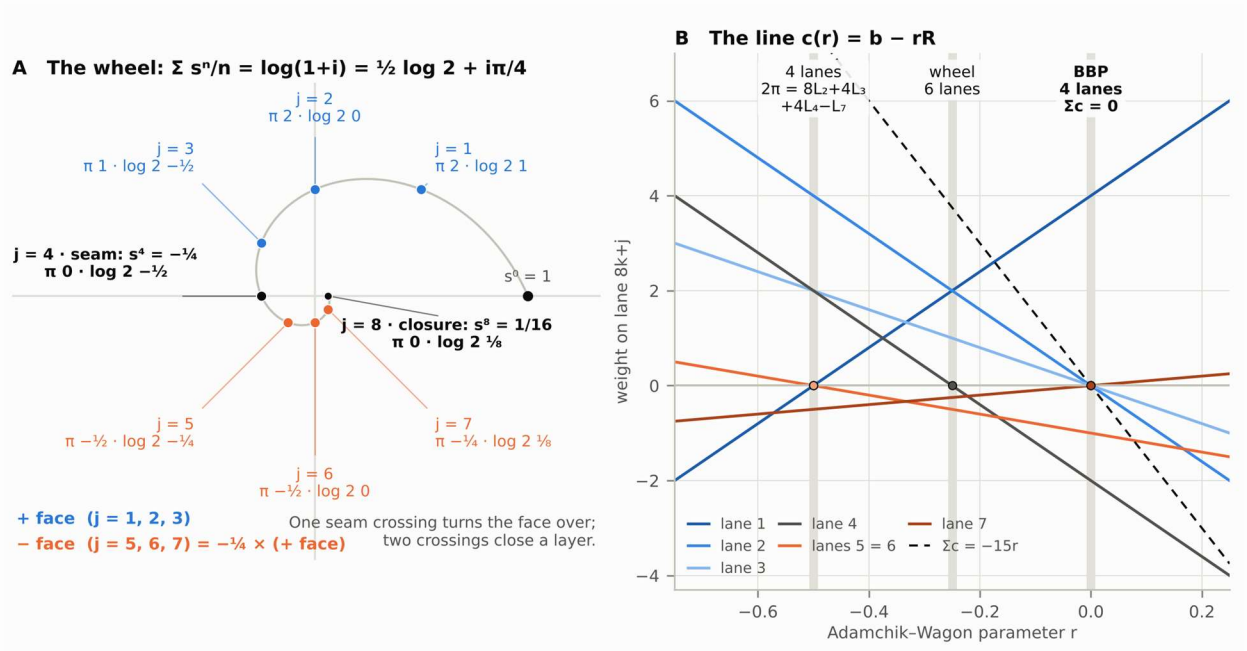


Figure 2. (A) The wheel $s = (1+i)/2$. Lane j sits at s^j; its π weight is 4 Im s^j and its log 2 weight 2 Re s^j. The face j = 1, 2, 3 is read positively; the seam ($j = 4$, $s^4 = -¼$) turns it over; the face j = 5, 6, 7 is the first face times −¼; j = 8 closes the layer at 1/16. (B) The Adamchik–Wagon line $c(r) = b - rR$. Several weights meet zero together only at r = 0 (BBP), r = −¼ (the wheel) and r = −½; the dashed line $\Sigma c = -15r$ vanishes only at BBP.

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7 Retain: the Mold at every scope

The Mold is the construction $G(a, b) = (b, F(b) \star a)$, with inverse $G^{-1}(b, c) = (F(b)^{-1} \star c, b)$, valid for any map F and any group operation $\star$. It is Fredkin's second-order technique for reversible cellular automata [9, 12]: carrying the prior state makes any update reversible. In the BBP machine it appears at every scope, always with the same shape.

Proposition 8 (one law for every register). *The phase ($x' = 16x - q$), the layer carry ($J' = 16J + w - H$, by (6)) and the lift ($S' = 16S - C$, Proposition 2) all obey new = 16·old − emitted, with an integer emitted at each step. In every case old = (new + emitted)/16.*

Theorem 9 (where the information goes). *(a) For odd d, $r \mapsto 16r \bmod d$ permutes $\mathbb{Z}/d$, so an odd lane's crossing is determined by its next state: the lane is reversible without retention. (b) An even lane ($d = 40$ or 20, o odd) needs retention only at its first step: its first state is the sealed residue 1, and every later state lies on its loop, where multiplication by 16 is invertible. Hence, given n, the field of lane states at step n determines the field at every earlier step. (c) The phase map of π, $x \mapsto \{16x\}$, is 16-to-1, and the emitted digit is never determined by the next phase. With the retained digits as a second coordinate the motion becomes the baker's map $(x, y) \mapsto (\{16x\}, (\lfloor 16x \rfloor + y)/16)$, a bijection of $[0, 1)^2$, the natural extension of $x \mapsto \{16x\}$ [11]. The hexadecimal expansion of π is precisely the Mold coordinate that the projection from lanes to π erases.*

Proof. (a) $\gcd(16, d) = 1$. (b) Write $d = 2^e o$ with $e \in \{1, 2\}$; then $16^m \equiv 0 \pmod{2^e}$ for $m \geq 1$, and 16 is invertible modulo o, so the state at age $m \geq 1$ is fixed by its residue modulo o (Chinese remainder theorem). (c) Each x has the sixteen preimages $(q + x)/16$, $q = 0, \ldots, 15$. The baker's map is inverted by $q = \lfloor 16y' \rfloor$, $y = \{16y'\}$, $x = (q + x')/16$. ■

We ran 2,404 lanes forward 600 steps and back to step 0 using only the field state and the step count; every lane was recovered exactly at every step. So the lanes lose no information, and π's phase does. A machine that keeps only π's phase must carry the Mold inside itself; Section 10 measures the price for the one-point machine, one hex digit of precision per step.

8 Invert at the scale of the loop: zero needs an inversion

A lane $d = 2^e o$ (o odd, $e \leq 2$) runs, after at most one step, on a loop

$$\{\, y/o : y \in aH \,\}, \quad H = \langle 16 \rangle \subset (\mathbb{Z}/o)^\times, \tag{13}$$

with $a = 1$ for $j = 1$ and 5, $a = 4$ for $j = 4$, and $a = 8$ for $j = 6$, because $16^m/(40) = 4 \times 16^{m-1}/o$ and $16^m/(20) = 8 \times 16^{m-1}/o$. Let μ_d be the mean of y/o over the loop ($\mu_d = 0$ for the degenerate lanes $d = 1$ and $d = 4$). Summing $16x = q + x'$ over a period shows that the lane's mean crossing is $15\mu_d$. We call a lane *balanced* when $\mu_d = ½$, mean crossing 7.5.

Theorem 10 (character law). *$\mu_d - ½ = (1/\varphi(o)) \Sigma \chi(a^{-1}) S(\chi)$, the sum running over the odd Dirichlet characters χ mod o with $\chi(16) = 1$, where $S(\chi) = (1/o) \Sigma_y \chi(y)\, y$, summed over $1 \leq y < o$, equals $B_{1,\chi^*} \prod_{p|o} (1 - \chi^*(p))$ and χ^* is the primitive character inducing χ.*

Proof. By orthogonality the indicator of aH is $(|H|/\phi(o)) \Sigma_{\chi(H)=1} \chi(a^{-1})\, \chi(y)$. The trivial character contributes ½. An even character contributes nothing, since replacing y by $o - y$ gives $2\Sigma\chi(y)y = o\Sigma\chi(y) = 0$. The

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product is the standard relation between the generalized Bernoulli number of an imprimitive character and that of its primitive form [13, Ch. 4]. ■

We checked the law against direct loop means on all 488 lanes with $k < 400$ and prime $o \geq 5$ (largest discrepancy 2.4×10^{-14}), and the product form on composite moduli such as $o = 1{,}921 = 17 \times 113$.

Corollary 11 (class numbers). *If $o > 3$ is prime, $o \equiv 3 \pmod 4$, and $\langle 16 \rangle$ is the subgroup of squares (equivalently, $\mathrm{ord}_o(2)$ is $o - 1$ or $(o - 1)/2$), then $\mu_d = \tfrac{1}{2} - (a/o) \cdot h(-o)/(o - 1)$, where (a/o) is the Legendre symbol and $h(-o)$ the class number of $\mathbb{Q}(\sqrt{-o})$.*

Only the quadratic character survives in Theorem 10, and $B_{1,\chi} = -h(-o)$ for it. Examples: $d = 14$ ($o = 7$) runs on $\{1/7, 2/7, 4/7\}$ with $\mu = \tfrac{1}{3} = \tfrac{1}{2} - h(-7)/6$; $d = 22$ ($o = 11$, $a = 8$, $(8/11) = -1$) runs on $\{16, 14, 4, 20, 12\}/22$ with $\mu = 0.6 = \tfrac{1}{2} + h(-11)/10$. Lanes $8k + 6$ with prime $o \equiv 3 \pmod 8$ lean high; every other lane covered by the corollary leans low. This is the base-16 lane analogue of Girstmair's formula for the digit sum of the period of $1/p$ [7, 8].

Theorem 12 (zero needs an inversion). *Let $o > 1$ be odd, $H = \langle 16 \rangle \subset (\mathbb{Z}/o)^\times$, and let μ be the mean of y/o over a coset aH.*

(a) If $-1 \in H$, then $\mu = \tfrac{1}{2}$ (Midy [10]; see also [6]).

(b) If $|H|$ is odd, equivalently if H contains no involution, then $\mu \neq \tfrac{1}{2}$.

(c) Otherwise H contains exactly one involution $\varepsilon \neq -1$, a partial inversion*: $o = o_1 o_2$ with $o_1, o_2 > 1$, $\varepsilon \equiv 1 \pmod{o_1}$ and $\varepsilon \equiv -1 \pmod{o_2}$. Write $y/o \equiv \alpha(y) + \beta(y) \pmod 1$ with $\alpha(y) \in (1/o_1)\mathbb{Z}$ and $\beta(y) \in (1/o_2)\mathbb{Z}$ in $(o, 1)$. Then*

$$\mu - \tfrac{1}{2} = \mathrm{E}[\alpha] - \mathrm{P}(\beta < \alpha), \tag{14}$$

expectation and probability taken over the coset. Balance occurs exactly when the share of the coset with $\beta < \alpha$ equals the mean of α.

(d) Every involution of $(\mathbb{Z}/o)^\times$ is ± 1 on each prime-power factor of o, so (a)–(c) cover all cases. When o is a prime power, case (c) cannot occur, and $\mu = \tfrac{1}{2}$ if and only if $-1 \in H$.

Proof. (a) $y \mapsto o - y$ maps aH onto itself. (b) The coset sum Σy is an integer, but balance requires it to equal $|H|o/2$, which is not an integer when $|H|$ and o are odd; a cyclic group has an involution exactly when its order is even. (c) ε fixes α, sends β to $1 - \beta$, and acts on aH without fixed points. From $\{\alpha + \beta\} = \alpha + \beta - [\alpha + \beta \geq 1]$ and this pairing, $\mathrm{E}[\beta] = \tfrac{1}{2}$ and $\mathrm{P}(\alpha + \beta \geq 1) = \mathrm{P}(\beta \leq \alpha)$; $\beta \neq \alpha$ because their denominators are coprime and neither is o. (d) $(\mathbb{Z}/p^e)^\times$ is cyclic for odd p. ■

Corollary 13. *Lanes $8k + 5$ and $8k + 6$ never contain the full inversion. They balance only through partial inversions, and they do: the first cases are $d = 493$ ($o = 17 \times 29$) and $d = 582$ ($o = 3 \times 97$).*

Proof. If $-1 \in \langle 16 \rangle \bmod o$, the same holds mod every prime p dividing o. For $j = 6$, $o = 4k + 3$ has a prime factor $p \equiv 3 \pmod 4$, since a product of primes $\equiv 1 \pmod 4$ is $\equiv 1 \pmod 4$; -1 is not a square mod p, while 16 is. For $j = 5$, $o \equiv 5 \pmod 8$ has a prime factor $p \not\equiv 1 \pmod 8$ for the same reason; -1 is a fourth power mod p only when $p \equiv 1 \pmod 8$, while every element of $\langle 16 \rangle$ is a fourth power. ■

Table 2 classifies every lane with $k \leq 128{,}000$, and formula (14) was confirmed in exact arithmetic on all 804 partial lanes with $k \leq 2{,}000$. A natural guess, that lanes $8k + 5$ and $8k + 6$ never balance, is false: 553 and 710 of them do. The guess confuses "no full inversion" with "no inversion". Figure 3 shows the four

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cases on small lanes. For d = 476 (o = 119 = 7 × 17, ε = 50) the loop {4, 64, 72, 81, 106, 30}/119 has mean exactly ½ although it does not contain its mirror image: $E[\alpha] = P(\beta < \alpha) = \frac{2}{3}$.

Table 2. Lanes with k ≤ 128,000 (modulus o > 1) by the involutions in ⟨16⟩; balanced lanes in parentheses.

Lane class	No involution	Full inversion (−1 ∈ H)	Partial inversion only
8k + 1	104,823 (0)	7,085 (7,085)	16,092 (656)
8k + 4	109,331 (0)	1,994 (1,994)	16,675 (812)
8k + 5	109,740 (0)	0	18,261 (553)
8k + 6	110,029 (0)	0	17,972 (710)
All	**433,923 (0)**	**9,079 (9,079)**	**69,000 (2,731)**

So the Form's "zero requires inversion" holds, in a sharper form than expected. Exact zero is impossible without an involution, forced by the full inversion, and only permitted by a partial one, where it becomes a statement about equidistribution between the fixed and the reflected parts of the loop.

Where the inversion comes from. In this machine the inversion is never imported. If a motion g first returns to the identity after 2m steps, then g^m squares to the identity without being it, so g^m is an involution; a motion of odd period has none. In the wheel the unit turn has period 8 and its half-period power is −1: the seam $s^4 = -¼$ is the midpoint of the layer's closure $s^8 = 1/16$, not an added rule. In a lane, 16 has period |H| on the loop, and when |H| is even its half-period power $16^{|H|/2}$ is exactly the involution ε of Theorem 12, full or partial. So zero needs an inversion, and an inversion needs an even closure: of the 512,002 lanes with k ≤ 128,000, 78,079 have even loops, and every balanced lane is among them.

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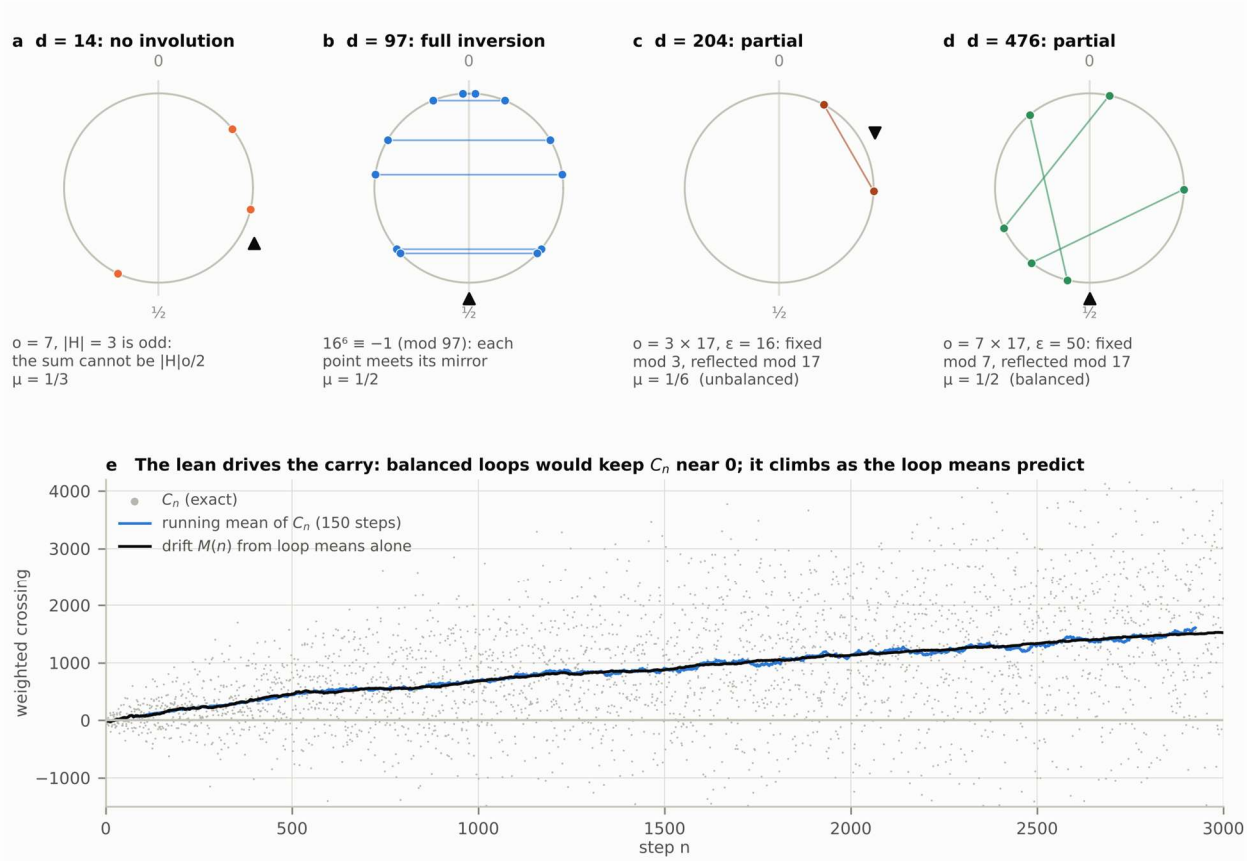


Figure 3. Zero needs an inversion. Each circle is a phase circle (o at the top, ½ at the bottom; the mirror $x \mapsto 1 - x$ is reflection in the vertical axis). Dots are a lane's loop, the black triangle marks its mean phase μ, and chords join each point to its partner under the loop's involution. (a) No involution: $\mu \neq ½$ by parity. (b) Full inversion: $\mu = ½$ (Midy). (c) Partial inversion, unbalanced. (d) Partial inversion, balanced, as permitted by (14). (e) The weighted crossing C_n of the whole field (gray), its running mean (blue), and the drift M(n) of (15) computed from loop means alone (black).

9 What the lean does: the drift of the carry

Because $\Sigma c_j = 0$, balanced loops would keep the weighted crossing C_n near zero on average. The loops lean, mostly below ½: for $k \le 1{,}000$ the mean of $\mu - ½$ is −0.019, −0.035, −0.025 and −0.027 in the classes $8k + 1$, $8k + 4$, $8k + 5$ and $8k + 6$, and with weights (4, −2, −1, −1) the net effect is upward. Replacing every lane's crossing by its loop mean $15\mu_d$ gives the mean-field drift

$$M(n) = 15 \sum_{k \le n} \sum_j c_j \, \mu_{8k+j}. \qquad (15)$$

Observation 14 (the loops predict the drift). *For $n < 3{,}000$ the window means of the exact C_n agree with those of M(n) to within 0.2–1.6% (Table 3), although single values of C_n fluctuate with standard deviation 400 to 1,200 (Fig. 3e).*

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Table 3. The carry and its mean-field prediction.

Window of n	Mean of C_n (exact)	Mean of M(n)	Ratio
[0, 500)	218.0	217.3	1.003
[500, 1,000)	560.9	552.0	1.016
[1,000, 2,000)	918.4	910.5	1.009
[2,000, 3,000)	1,338.4	1,335.2	1.002

Observation 15 (growth). *M(n) = 680.7, 1,885.5, 4,514.8, 10,276.3 and 15,391.1 at $n = 10^3$, 4×10^3, 1.6×10^4, 6.4×10^4 and 1.28×10^5. The local exponent falls from 0.73 near $n = 10^3$ to 0.58 near 10^5, and $M(n)/(\sqrt{n} \ln n)$ stays within [3.60, 3.75] for $1.6 \times 10^4 \le n \le 1.28 \times 10^5$.*

The lean of a typical lane is of order $o^{-1/2}$, since class numbers are of order $\sqrt{o}$ on average, so a law of the form $\sqrt{n}$ times a slowly varying factor is what one expects; we do not prove one. An exponent of about 0.8 fitted on $n \le 3{,}000$ is a small-n effect. The drift is absorbed by the integer part of the lift: by (4), $16I_n - I_{n+1} = C_n - \delta_{n+1}$, and $|I_n|$ stays below 336 for $n < 3{,}000$.

10 Read: the single traveler and its gap

Bailey and Crandall [5] fed the layers to a single point,

$$x_0 = 0, \quad x_n = \{16\, x_{n-1} + r(n)\}, \tag{16}$$

$$r(n) = V_{n-1} = (120n^2 - 89n + 16) / (512n^4 - 1024n^3 + 712n^2 - 206n + 21),$$

and conjectured (their Conjecture 3.1) that $\lfloor 16x_n \rfloor$ is the n-th hexadecimal digit δ_n of π for every n, having checked 100,000 digits. This traveler has no lanes, no stored digits and no π in its loop. It is the machine with everything compiled into one register.

Theorem 16 (misread criterion). *Let $T_n = \Sigma_{j \ge 0}\, V_{n+j}\, 16^{-j} = 1/(4n^2) + O(n^{-3})$. Then $\lfloor 16x_n \rfloor \ne \delta_n$ if and only if $\{16^n \pi\} < T_n$, and in that case $\lfloor 16x_n \rfloor \equiv \delta_n - 1 \pmod{16}$.*

Proof. Unrolling (16) and using (2), $x_n = \{16^{n-1}\pi - T_n/16\}$. Put $\phi = \{16^{n-1}\pi\} = 0.\delta_n\delta_{n+1}\ldots$ in base 16. Since $0 < T_n/16 < 1/16$, subtracting it changes the digit at position n exactly when it borrows from that position, that is, when $16\phi - \delta_n = \{16^n\pi\} < T_n$; the borrow lowers the digit by one, modulo 16. The asymptotic form of T_n follows from (5). ■

In words: the traveler misreads only where π's own digits after position n open with a run of zeros long enough to hide the pushes it has not yet received, about $\log_{16}(4n^2)$ zeros. Equivalently, Conjecture 3.1 says that $\pi - p/16^n \ge T_n\, 16^{-n}$ for $p = \lfloor 16^n\pi \rfloor$ and every $n \ge 1$, a one-sided Diophantine inequality with denominators 16^n.

We computed 4,000,040 hexadecimal digits of π independently (mpmath) and evaluated the criterion at every $n \le 4 \times 10^6$: there is no misread. The closest approach is at n = 1, where $\{16\pi\}/T_1 = 2.009$; no other n has ratio below 3. If the tails $\{16^n\pi\}$ were uniformly distributed, the expected number of misreads over all

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$n \geq 1$ would be $\Sigma T_n \approx 0.261$, and beyond $n = 4 \times 10^6$ about 6.2×10^{-8}. Large-scale statistical studies of π's hexadecimal digits [4] are consistent with that heuristic.

What is proved falls far short. The irrationality measure of π is at most 7.1032 [14] (7.1019 in a recent preprint [15]). It gives $|\pi - p/16^n| > 16^{-(\mu(\pi)+\varepsilon)n}$ for large n, which excludes zero runs longer than about $(\mu(\pi) - 1)n \approx 6.1n$. Conjecture 3.1 needs runs of length about $2 \log_{16} n$ excluded. The distance between a linear and a logarithmic bound is the whole gap, and no current Diophantine method narrows it. Since $x_n = \{16^{n-1}\pi - T_n/16\}$ with $T_n \to 0$, equidistribution of (x_n) is equivalent to the normality of π in base 16 [5].

The traveler also pays for carrying the Mold inside one register. In floating point it loses one hex digit of precision per step: with 16, 50 and 100 decimal digits it stays exact for 13, 42 and 83 steps, which is $\log_{10}16 \approx 1.204$ decimal digits per step, as Theorem 9(c) predicts for a 16-to-1 map.

11 Discussion: one Form, every scope

Table 4 collects the correspondence. Three things recur. The same law, new = 16·old – emitted, runs the phase, the layer carry and the lift. The same inversion governs the weights ($\Sigma c = 0$ gives the crest and the silent band) and the loops (an involution is needed for balance), and in the wheel and the loops alike it arises as the half-period of an even closure. The same seam relates base −4 and base 16, and the same wheel holds π and log 2 as its two shadows, with one kernel vanishing at its first step. That is what "the same Mold at every scope" means in this machine, and none of it requires leaving the formula.

Table 4. The Form in the BBP machine.

Form	Where it acts	Statement
Compile first	The field at step 0	All terms present; $S_0 = \pi - 4$ contains the unborn tail (Prop. 1)
Nest	A lane before its first crossing	Phase grows ×16 without crossing (Prop. 3)
Run	First crossing	Forced at step $k + \lceil \log_{16} d \rceil - 1$ (Prop. 3)
Encapsulate	Four lanes into one layer	Value arrives near $\log_{16}(64k^2/15)$; carries before it are value-free (Thm. 4)
Invert	Weights; loops	$\Sigma c = 0 \Leftrightarrow$ crest $\Leftrightarrow$ silent band (Thm. 5); balance needs an involution, the half-period of an even loop (Thm. 12)
Fold	Wheel against its mirror	log 2 is the fold, π the inversion (Thm. 6b); $s + \bar{s} = (½ + ½) + (½ - ½)i = 1$ (Cor. 7)
Seam	Half-turn $s^4 = -¼$	Base −4 read across two seams is base 16 (Thm. 6c)
Retain (Mold)	Every register	new = 16·old – emitted; π's digits are the erased coordinate (Thm. 9)
Read	Crossing, digit, traveler	The difference is the gate (Prop. 2); misread iff a zero run (Thm. 16)

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Open problems

1. The Bailey–Crandall conjecture that the traveler never misreads. By Theorem 16 it is a one-sided Diophantine inequality with denominators 16^n and needs zero runs of length about $2 \log_{16} n$ excluded.
2. Normality of π in base 16, equivalent to the equidistribution of the traveler.
3. The asymptotic law of the drift M(n), and a proof that the exact C_n follows it.
4. Which partial-inversion cosets balance? About 4% of partial lanes with $k \le 1.28 \times 10^5$ do. Is there a density, and a structural description beyond (14)?
5. Whether the crest and the shared kernel of Corollary 7 organize BBP-type formulas for other constants.

We have not addressed the question that motivates the Form, whether a first difference compels inversion. The BBP machine shows the Form operating; it does not show where the Form comes from.

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Appendix: Reproducibility

All computations use Python 3 with mpmath, sympy and exact rational arithmetic, plus one C program. The scripts and their logs accompany this paper as bbp-form-scripts.zip; times are for one CPU core.

Table 5. Scripts.

Script	Verifies	Time
one_law.py 2000	Props. 1–3: the one-motion machine against digit extraction, 2,000 steps	35 s
layers.py	Thm. 4: identity (6), onsets, silent band, head and tail, variances	1 s
check/claims.py	Identity (4), Table 3, Obs. 15, $\|I_n\|$, $J_k(0) = 0$	7 s
form/form_checks.py	Thm. 5, Thm. 6, Cor. 7: crest, wheel, fold, seam, sparsity, shared kernel	1 s
form/mold_check.py	Thm. 9: lane field forward and back 600 steps	6 s
check/loops.py	Thm. 10, Cor. 11: character law, quadratic term	3 s
check/euler_zero.py	Thm. 10: product form on composite moduli	2 s
check/lean.c	Thm. 12, Cor. 13, Table 2; writes M(n) up to k = 128,000	80 s
check/partial.py	Thm. 12(c): formula (14) exact on 804 partial lanes	1 s
edge/one_point.py	The traveler (16) against π; precision cost	1 s
wheel/gap_scan.py	Thm. 16: the criterion for every $n \le 4 \times 10^6$	3 min
figs/fig_*_print.py	Figures 1–3	15 s

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